

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course:B.Tech

Branch: Computer Engg/Comp. Science &Engg(AI) & Allied

Semester : 3rd

Subject Code: BTCOC305A Object Oriented Programming in C++

Max Marks: 60

Date:21/02/2025

Duration: 3 Hr.

Instructions to the Students:

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include Objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Information hiding can be achieved through_____. a) Encapsulation, Inheritance b) Overloading b) Encapsulation, Abstraction d) Encapsulation, Polymorphism	CO1	1
2	Which of the following features of OOP is used to derive a class from another? a) Encapsulation b) Inheritance c) Data hiding d) Polymorphism	CO1	1
3	Which of the following may not be an integral part of an object? a) State b) behavior c) Protected data members d) All of given	CO1	1
4	What is a constructor? a) A class automatically called whenever a new object of this class is created. b) A class automatically called whenever a new object of this class is destroyed. c) A function automatically called whenever a new object of this class is created. d) A function automatically called whenever a new object of this class is destroyed.	CO1	1
5	Data items in a class must be private. Select correct option: a) True b) False	CO2	1
6	Suppose there is an object of type Person, which of the following can be considered as one of its attributes Select correct option: a) Name b) Age c) Work d) Both Name and Age	CO2	1
7	If MyClass has a destructor what is the destructor named? a) MyClass b) ~MyClass c) My~Class d) MyClass~	CO2	1
8	When class B is inherited from class A, what is the order in which the constructors of those classes are called a) Class A first Class B next b) Class B first Class A next C) Class B's only as it is the child class d) Class A's only as it is the parent class	CO2	1
9	Which of the following types of binding used in C++? a) Static binding b) Dynamic binding c) Pointer binding d) Inheritance binding	CO3	1

10	What will be the output of the following C++ code? <pre>#include <iostream> using namespace std; int main() { char c = 74; cout << c; return 0; }</pre> a) I b) J c) A d) N	CO3	1
11	Which of the following is used to create an output stream? a) of stream b) if stream c) ostream d) fstream	CO4	1
12	By default, what a program does when it detects an exception? a) Continue running b) Results in the termination of the program c) Calls other functions of the program d) Removes the exception and tells the programmer about an exception	CO5	1
Q. 2	Solve the following.		
A)	Define the following terms: i) Object ii) Class iii) Abstraction iv) Constructor v) Inheritance vi) Encapsulation	CO1	6
B)	Differentiate between: i) POP and OOP ii) Class & Object	CO3	6
Q.3	Solve the following.		
A)	Elaborate any five special characteristics of a constructor.	CO3	6
B)	Explain the five forms of inheritances with neat diagram.	CO3	6
Q. 4	Solve Any Two of the following.		12
A)	What is polymorphism? Explain its types with neat diagram.	CO3	6
B)	Explain virtual function with suitable example.	CO3	6
C)	What is the difference between a friend function and a member function?	CO3	6
Q.5	Solve Any Two of the following.		
A)	Compare compile time polymorphism and run time polymorphism with suitable real time application.	CO4	6
B)	What is the application of this pointer? Explain with help of C++ program.	CO4	6
C)	Write a C++ program to declare class mobile having data members as price and model number. Accept and display this data for five objects.	CO4	6
Q. 6	Solve Any Two of the following.		
A)	What is an exception? How is an exception handle in C++ ? What are the advantages of using exceptions handling mechanism in a program?	CO5	6
B)	What are the components of STL. Explain In detail.	CO5	6
C)	Write the following any two short notes. (i) Iterators (ii) Container adaptors (iii) Vector	CO5	6
*** End ***			